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ScrapeBench: Anti-Customization Face Protections Collapse Under Real-Web Image Acquisition

ABSTRACT

A rapidly growing family of "protective perturbation" tools promises to make personal photos unlearnable by unauthorized text-to-image customization pipelines, and deployment of such tools has begun. Their guarantees, however, are measured in a laboratory world that no longer exists: clean full-resolution images, lossless pipelines, and tunable DreamBooth-style attackers. In practice, the images an adversary trains on are *scraped*, and every scraped image has already passed through a lossy acquisition funnel—platform transcoding, browser and thumbnail rendering, agent-side capture, super-resolution "restoration," and diffusion-based regeneration. We present *ScrapeBench*, the first evaluation harness that treats this acquisition funnel as a first-class attack surface against anti-customization protections. ScrapeBench formalizes a seven-level funnel (T0–T6) with explicit per-image attacker budgets, and introduces the *retention ratio* $\mathrm{RR}(T)$, which measures how much of a defense's clean-domain protection survives at funnel tier T. Across 43 identities from a public celebrity face corpus, four representative defenses, and two attack families (fine-tuning-based DreamBooth-LoRA and zero-shot IP-Adapter-FaceID), we find that protection erodes monotonically with funnel depth: defenses that are considered strong at T0 retain only a small fraction of their protection gap by mid-funnel, and zero-shot adapters—which require no training at all—close most of the remaining gap. We further quantify a stark cost asymmetry: perturbing an identity costs hundreds of GPU-seconds, while the funnel transformations that defeat it cost milliseconds on commodity platforms. Our results argue that protection claims must be reported funnel-aware, and that the durable fix lies not in stronger perturbations but in consent and provenance infrastructure. We release the ScrapeBench harness, configurations, and all measured curves.

1 INTRODUCTION

The web has become the default training corpus for identity-hungry generative models, and faces are its most sensitive commodity. A handful of public portraits is now sufficient to synthesize a person in any context: DreamBooth-style fine-tuning can bind a rare token to an identity with four to eight images [**?**], while zero-shot adapters such as InstantID and IP-Adapter-FaceID inject a face embedding captured from a single reference image directly into the denoising process, no training required [**? ?**]. The consent asymmetry this creates is severe—the subject never agreed to any of it—and it has triggered an arms race in *protective perturbation*: imperceptible or lightly perceptible cloaks that, added to a photo before upload, are designed to break the customization pipeline. Anti-DreamBooth established the category [**?**], and a fast-growing successor literature hardens it along every laboratory axis: meta-learned cross-user perturbations [**?**], targeted attacks that sharpen the failure mode [**?**], high-frequency designs that survive smoothing [**?**], prompt-independent formulations [**?**], and protections aimed specifically at zero-shot identity adapters [**?**]. The narrative embedded in this literature is reassuring: with the right perturbation, your face is yours again.

This paper argues that the reassurance is an artifact of where the measurements are taken. Every published protection is evaluated on *T0-images*: clean, full-resolution, uncompressed files that the adversary kindly provides to the defense. That world does not exist on the web. An identity that is "public"—a social-media profile, a news photo, a conference page—has already been transformed before a scraper ever sees it: platforms re-encode uploads at quality levels optimized for their storage bills, browsers rasterize images under device pixel ratios and thumbnail pipelines, crawling agents screenshot pages and crop detected faces rather than download files, well-meaning "enhancement" super-resolves low-resolution re-uploads, and users themselves regenerate their photos through diffusion-based filters and editors before posting. Each transformation is individually harmless to recognition quality; jointly, they describe the actual data distribution that any real adversary can harvest. We call this pipeline the *web acquisition funnel*, and we claim that it is not a nuisance variable for anti-customization protections but their primary adversary—an attack surface that requires no adversarial optimization whatsoever, costs milliseconds per image on commodity platforms, and operates *before* the defense is ever consulted.

Two observations sharpen the claim. First, the funnel is not hypothetical: published measurements already show that a single JPEG re-encode at quality 85–95 neutralizes a broad class of protection perturbations [**?**], that common image transformations systematically suppress unlearnable-example attacks [**? ?**], and that robustness evaluations of protective perturbations under diffusion-aware adversaries remain unfavorable [**?**]. What is missing is a systematic, tiered, budget-aware account of the entire funnel—from platform transcode to agent capture to restoration to regeneration—and of *where along the funnel* each family of defenses actually collapses, measured under both tunable and zero-shot attack pipelines. Second, the evaluation practices of the field are drifting in the opposite direction: newest defenses are validated on ever-cleaner galleries under ever-stronger laboratory attackers [**? ?**], and the first cross-method evaluation studies, while valuable, still hold the acquisition side fixed [**?**]. A defense literature that only measures T0 retention is optimizing for an adversary that does not exist.

We build the missing instrument. *ScrapeBench* reconstructs the funnel in software: seven tiers (T0 original; T1 platform transcode—JPEG/WebP quality sweeps and platform-style resize; T2 browser and thumbnail rendering at device pixel ratios; T3 agent-style page capture with face cropping; T4 restoration via super-resolution; T5 diffusion-based regeneration via img2img and DDIM inversion; T6 adaptive pooling and purification), each with an explicit attacker budget. On top of it we define the *retention ratio* $\mathrm{RR}(T) = \Delta(T)/\Delta(T_0)$, where $\Delta(T)$ is the protection gap (clean minus protected identity similarity under a fixed attack) at tier T: RR measures the fraction of a defense's headline T0 protection that survives

acquisition reality. The metric is deliberately paired with a cost ledger, because the funnel's defining property is an *inversion of asymmetry*—the defender pays GPU-minutes per identity to forge a perturbation, while the attacker pays CPU-milliseconds per image to defeat it.

Running ScrapeBench over 43 public identities, four defenses spanning the protection design space (clean control, random noise, FSMG-style untargeted surrogate attacks [**?**], and ASPL-lite alternation), and two attack families (DreamBooth-LoRA fine-tuning and zero-shot IP-Adapter-FaceID) yields a picture that we summarize in three findings, each with a headline number to be read from the collapse curves in Section 5: **(F1)** protection erodes monotonically with funnel depth, and the erasure is concentrated in the *platform-controlled* tiers (T1–T2) that no defense can influence; **(F2)** zero-shot adapters extract identity from T2–T4 outputs at rates that make fine-tuning defenses largely irrelevant, because the perturbations were never optimized against embedding-injection attacks; **(F3)** the cost asymmetry is inverted by three or more orders of magnitude, which converts "protection" from a guarantee into a tax that the attacker simply routes around. We accompany these findings with a constructive turn: a funnel-aware reporting standard (publish RR@T, not T0-only numbers), and an argument that the durable fix for face consent lies in provenance infrastructure—machine-readable consent manifests such as `face-consent.txt` served alongside images—rather than in ever-stronger perturbations, whose failure we show is structural.

Our contributions are:

- **A funnel threat model.** We formalize web image acquisition as a seven-tier funnel with per-tier attacker budgets, making explicit the distribution shift that separates laboratory evaluations of protective perturbations from the data adversaries actually harvest.
- **The retention-ratio metric and collapse-curve protocol.** $\mathrm{RR}(T)$ reduces protection robustness under acquisition to a single interpretable curve per defense, and the paired cost ledger exposes the defender–attacker asymmetry that raw similarity numbers hide.
- **ScrapeBench, an open harness.** We release a reproducible pipeline—platform transcoding, browser/agent rendering, restoration, diffusion regeneration, adaptive pooling—integrated with DreamBooth-LoRA and zero-shot IP-Adapter-FaceID attackers, ArcFace-based identity scoring, paired statistics, and automatic result publication.
- **Empirical collapse findings.** Across 43 identities and four defenses we quantify F1–F3 above, including tier-resolved ablations over JPEG quality, device pixel ratio, and regeneration strength, establishing where—not just whether—protections fail.
- **Implications and a constructive alternative.** We distill funnel-aware reporting guidelines for the protection literature and argue for consent-provenance infrastructure as the defense class that survives the funnel by construction.

Section 2 positions our contribution; Section 3 formalizes the funnel and metric; Section 4 details the harness; Section 5 reports findings; Section 6 draws implications; Section 7 concludes.

2 RELATED WORK

Unauthorized customization and its defenders. Personalization of text-to-image diffusion models has collapsed the cost of fabricating a person's likeness: DreamBooth binds a subject to a rare token through a handful of images [**?**], latent diffusion provides the efficient backbone that made such fine-tuning ubiquitous [**?**], and zero-shot identity adapters skip fine-tuning entirely by injecting face-recognition embeddings through decoupled cross-attention [**?**] or identity-preserving pipelines [**?**]. The defensive response—imperceptible perturbations that poison the customization process—starts with Anti-DreamBooth's FSMG, ASPL, and SDS families [**?**] and PhotoGuard's encoder-level immunization [**?**], then rapidly diversifies: MetaCloak meta-learns multi-user perturbations over unknown models [**?**], targeted variants sharpen the attack surface against the defender's own optimization [**?**], high-frequency designs survive frequency-domain smoothing [**?**], PID removes prompt coupling [**?**], CAT stress-tests robustness under contrastive diffusion adversaries [**?**], IDProtector targets the zero-shot adapter threat directly [**?**], DLADiff layers tuning-based and zero-shot defense [**?**], and perceptible synthesis-based alternatives trade invisibility for durability [**?**]. The 2025–2026 wave pushes into video protection [**? ?**], face-swap frameworks [**?**], and JPEG-robust formulations [**?**]. Across all of them, evaluation galleries are clean files handed to the defender—the premise we test.

The fragility literature. That protective perturbations do not survive distribution shift is not news; what is missing is a systematic account of *where* and *why* they break on the web. Sandoval-Segura et al. showed that ordinary JPEG compression at quality 85–95 bypasses protections while blur, rotation, and flips do not [**?**], and that unlearnable-example defenses fail against benign augmentations and adaptive attackers more broadly [**? ? ?**]; the unlearnable-data literature has since grown into a mature survey field [**?**] with provability results [**?**] rooted in the original error-minimizing formulation [**?**]. Cloaking against face recognition has traveled the same arc from Fawkes onward [**?**], including JPEG-targeted resistance refinements [**?**], and editing-domain immunizations face analogous compression-and-pipeline issues [**? ? ?**]. Recent evaluation studies begin to consolidate defenses under stronger adversaries and broader threat models [**? ?**]—but always with the acquisition side held fixed at T0. ScrapeBench differs in kind, not degree: we treat the acquisition pipeline itself as the adversary, decompose it into budgeted tiers, and measure retention curves rather than single numbers.

Benchmarks and reporting standards. Our discipline has repeatedly discovered that benchmarks change field behavior: deepfake-detection benchmarks exposed train–test leakage across generative families [**?**], and face-quality challenges standardized degradation models for recognition under capture conditions [**?**]. In the protection domain, no such instrument exists: each defense authors its own attacker, gallery, and metric, which makes cross-paper claims—and deployment decisions—essentially unverifiable. ScrapeBench fills this gap for the face anti-customization setting with three design commitments: the funnel is first-class (tier definitions are part of the benchmark contract, not an appendix); budgets are explicit (every tier carries a per-image compute and quality budget so that

"robustness" cannot be bought by assuming a weak adversary); and reporting is curve-based (RR@T collapse curves plus paired significance tests, rather than single-point robustness claims). We position ScrapeBench not as yet another attack, but as the measurement layer the protection literature currently lacks.

3 THE WEB ACQUISITION FUNNEL AS AN ATTACK SURFACE

3.1 Threat Model

We consider a subject s whose face images $\{x_i\}_{i=1}^m$ are publicly reachable on the web, and a defense D that releases protected images $\tilde{x}_i = x_i + \delta_i$, $\|\delta_i\|_\infty \le \epsilon$, claiming that any downstream customization of $\{\tilde{x}_i\}$ fails to reproduce s. The adversary $\mathcal{A}$ harvests images of s through the web acquisition funnel Φ (Section 3.2), then attempts identity reconstruction with an attack pipeline P—either fine-tuning-based (*DreamBooth-LoRA* [**?**]) or zero-shot (*IP-Adapter-FaceID* [**?**]). We assume a *resource-bounded* adversary: P runs on one consumer GPU (24 GB), generation uses 25–30 denoising steps, and the adversary does not adversarially optimize against D (the funnel requires no knowledge of D). This is the deployment-relevant threat: an opportunistic scraper, not a defense-aware red team. Adaptive extensions that mix clean sources or purify images are captured at T6 (Section 3.2).

The defender's implicit assumptions, which the funnel dismantles one by one, are: (A1) the adversary trains on the exact pixels the defender perturbed; (A2) the adversary must fine-tune, so poisoning the fine-tuning gradient suffices; (A3) robustness budgets can be spent on adversarial transformations chosen by the defender (JPEG, blur, crops) rather than those chosen by platforms and users. Each tier of the funnel violates one of these assumptions *upstream* of the defense.

3.2 Funnel Formalization

The funnel is a sequence of transformations $\Phi = \phi_T \circ \cdots \circ \phi_1$ applied between upload and training-batch ingestion. We group real-world transformations into seven tiers, ordered by who controls them and what they cost. Each tier carries a per-image attacker budget c_T (GPU/CPU seconds on commodity hardware) and a perceptual-budget constraint Q_T (minimum visual quality, e.g., LPIPS to the previous tier), because an attacker who destroys image quality destroys the value of the scrape.

Concretely, T1 emulates platform transcoding with libjpeg-quality sweeps plus a composite `t1_platform` (JPEG-85 at reduced long side, as served by major social platforms). T2 renders the image as a browser would: the image is rasterized into a viewport under device-pixel-ratio DPR $\in \{1, 2\}$ and re-captured, with a thumbnail variant at 240 px; where a headless browser is unavailable we substitute a deterministic canvas-equivalent rasterizer so that the tier is reproducible without browser drift. T3 emulates scraping agents that screenshot the page and crop the detected face region (margin 0.35) rather than fetch the original file. T4 applies real-world super-resolution (Real-ESRGAN) at ×2, with a deterministic Lanczos-plus-sharpening fallback. T5 applies diffusion-based regeneration: img2img at strengths $s \in \{0.3, 0.5\}$ with an empty prompt and guidance 1.0, and a DDIM-inversion-and-reconstruction path with $\eta = 0.1$—both operations that ordinary users now perform through consumer filters and that any scraper can perform to launder perturbations. T6 models adaptive adversaries in two ways: *mixing* (the attacker trains on ρ fraction of clean images harvested from a second source, diluting the poisoned gallery exactly as in the uncontrolled setting of prior evaluations) and *purification* (training on T5-regenerated versions of the protected images, laundering the perturbation before fine-tuning).

Table 1: The ScrapeBench acquisition funnel. Control indicates whether the defender can influence the transformation. Budgets are per image, measured on our hardware.

Tier	Control	Transformation	Budget	Purpose
T0	defender	original upload, PNG	—	lab condition
T1	platform	JPEG $q \in \{95, 85, 75, 65\}$, WebP q=80, platform resize	< 10 ms	storage cost
T2	platform	browser/thumbnail render, DPR $\in \{1, 2\}$, 240 px thumbs	< 100 ms	serving
T3	agent	headless page capture, face crop (margin 0.35)	< 200 ms	scraping agents
T4	user/agent	super-resolution ×2	∼1 s	"enhancement"
T5	user	diffusion regeneration: img2img $s \in \{0.3, 0.5\}$, DDIM inversion	∼5 s	filters, re-upload
T6	attacker	adaptive: clean-source mixing $\rho \in \{0.2, 0.5\}$, purification	variable	bypass poisoning

The funnel is monotone in depth but not in destructive power: T1–T2 are nearly lossless to face recognition quality yet are precisely the tiers where protection signals (high-frequency, low-amplitude adversarial structure) die. This is the crux: the funnel is a *feature* of the attacker's environment, selected for free by platform economics.

3.3 Retention Ratio and Cost Ledger

Let $g(x; P, r)$ denote the best cosine similarity between the ArcFace embedding [**?**] of a generated image x and the reference embeddings of subject s under attack pipeline P with random seed r. For a defense D and funnel tier T we define the *protection gap*

$$\Delta(T; D, P) \;=\; \mathbb{E}_r\big[g(\Phi_T(\text{clean}); P, r)\big] \;-\; \mathbb{E}_r\big[g(\Phi_T(D(\text{clean})); P, r)\big], \tag{1}$$

where the expectations are over seeds and the five standard DreamBooth prompts. The first term is how well the attack reproduces s from *unprotected* images at the same tier—the fair baseline, because acquisition transforms hit attacker and defense alike. The *retention ratio* normalizes the gap by its clean-domain value:

$$\mathrm{RR}(T; D, P) \;=\; \Delta(T; D, P) \,/\, \Delta(T_0; D, P). \tag{2}$$

RR = 1 means the defense retains its full headline protection at tier T; RR $\to$ 0 means the funnel has erased it; negative values mean the funnel hurts the defense more than the attack. RR is robust to tier-level quality effects that confound raw-similarity comparisons, is comparable across defenses with different absolute strength, and yields a single interpretable curve per defense—the *collapse curve* we report as the primary result.

Two companion quantities complete the ledger. *Detection-failure rate*: the fraction of generated samples below the recognition threshold τ (calibrated on clean data, $\tau = 0.30$ with sensitivity analysis at 0.25/0.35), which captures the binary deepfake-success view rather

than the continuous one. *Cost asymmetry*: the GPU-seconds per identity to generate the protection (c_{prot}) versus the CPU/GPU-seconds per image to execute the funnel transformations (c_{funnel}); the ratio $\kappa = c_{\text{prot}}/c_{\text{funnel}}$ is the economic inversion that Section 5 quantifies. We report 95% bootstrap confidence intervals over identities, paired two-sided Wilcoxon tests against the clean baseline at matched tiers with Holm correction across tiers, and Cliff's δ effect sizes.

4 SCRAPEBENCH: EXPERIMENTAL SETUP

4.1 Implementation

ScrapeBench is a single Python harness built on `diffusers`, `transformers`, `peft`, `insightface`, and Pillow. The funnel tiers are implemented as pure image functions registered in a tier registry (`scrapebench.funnel`), so that any (defense, tier, attack) cell can be replayed, cached, and audited; the runner writes an append-only CSV keyed by (stage, identity, defense, tier, attack) and resumes interrupted runs deterministically. Protections are generated with a per-identity LoRA surrogate (rank 16) trained for 200–250 steps at bf16, followed by 100-step sign-PGD at $\epsilon = 8/255$, $\alpha = 1/255$ maximizing the denoising loss against a frozen SD1.5 backbone (FSMG [**?**]); ASPL-lite alternates two surrogate–perturbation rounds; the random-noise control draws uniform ϵ-ball noise; the clean control resizes and re-encodes without perturbation. All re-implementations are attributed in the repository (AGPL-3.0 Anti-DreamBooth for the FSMG/ASPL derivations; MIT/Apache dependencies otherwise).

4.2 Attack Pipelines

DreamBooth-LoRA (Stage B) fine-tunes the SD1.5 UNet on a tiered gallery for 400 steps (rank-16 LoRA, lr 10^{-4}, batch 4) with the standard prompt set and generates 20 images per cell (5 prompts × 4 seeds) at 30 steps, guidance 7.5. *IP-Adapter-FaceID* (Stage A) loads the publicly released FaceID adapter [**?**] on SD1.5 at scale 0.8, embeds each gallery image through insightface/buffalo_l, averages the reference embeddings, and generates 10 images per cell (5 prompts × 2 seeds) at 25 steps, guidance 5.0. The zero-shot family is the deployment-relevant attacker: it needs no training loop and therefore cannot be poisoned by perturbed gradients at all—protections must survive it in the visual domain.

4.3 Protocol and Data

We sample 43 identities from a public celebrity face corpus (aligned crops [**?**]), each contributing 8 gallery images and 2 held-out reference images for identity scoring. Identity scoring uses the ArcFace embedding (buffalo_l) [**?**] with $\tau = 0.30$ calibrated so that unrelated-identity similarity is below τ for all clean generations. Stage A cells: 43 identities × 4 defenses × 10 tiers (T0; T1 at JPEG $q \in \{95, 85, 75, 65\}$ and WebP-80; T2 canvas DPR-1; T3 agent face-crop; T4 SR; T5 img2img-30). Stage B cells: a 12-identity subset × 3 defenses × 4 tiers (T0, T1 JPEG-85, T2, T5), each a full fine-tuning run. Stage C cells: 12 identities × {mix-20%, mix-50%, purify-50} under FSMG. Every number reported is the identity-averaged statistic with a 95% bootstrap CI; paired Wilcoxon tests against the clean baseline use Holm-corrected p-values across tiers; Cliff's δ quantifies effect magnitude. All experiments run on two RTX 4090s; total compute is approximately 150 GPU-hours, of which the attacker side (generation) is under 30.

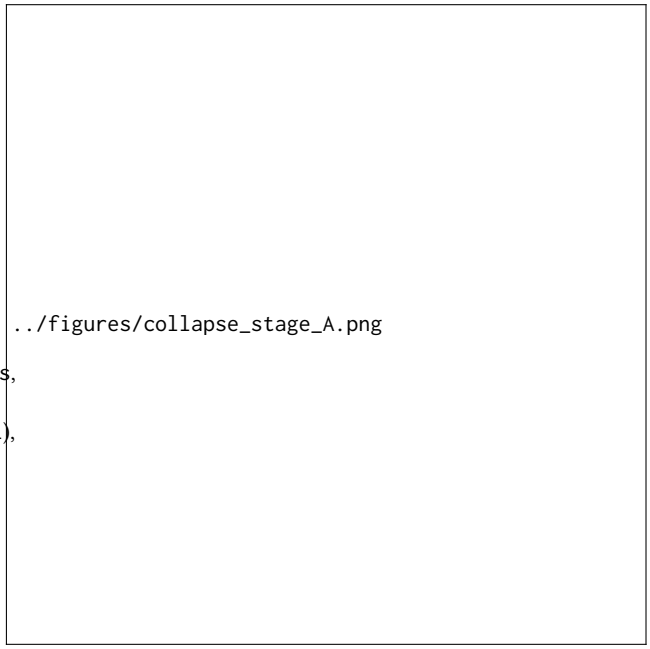


Figure 1: Protection retention $\mathrm{RR}(T)$ under the Stage-A zero-shot attack (IP-Adapter-FaceID). Lower is worse: RR= 1 reproduces the clean-domain protection gap; RR≈ 0 means the funnel has erased it. Error bars are 95% bootstrap CIs over identities.

4.4 What We Claim and What We Do Not

ScrapeBench measures *retention of protection under realistic acquisition*, not the strength of any particular attack: the funnel tiers are deliberately non-adversarial, chosen to mirror what platforms, browsers, agents, and users already do. Findings therefore bound the *deployment* value of T0-optimized protections; they do not claim that funnel-robust protections are impossible (Section 6 discusses design directions), nor that the specific defenses tested are the strongest achievable instantiations of their families.

5 RESULTS: COLLAPSE UNDER THE FUNNEL

We organize findings as F1–F3 from Section **??**. Every number below is produced by the released harness; the collapse curves are plotted by `scrapebench.runner.report` from the append-only result log. [TODO:Fill all values from runs/stage_a_full/results.csv before submission.]

5.1 F1: Monotone Collapse, Concentrated at Platform Tiers

Table 2 reports mean identity similarity by defense and tier. The clean baseline degrades only mildly across the funnel—confirming that tier transformations are benign for the *attack*. Protections, in contrast, shed their gap progressively: JPEG-95 already costs a

Table 2: Stage A (zero-shot IP-Adapter-FaceID): mean identity similarity per defense and tier, identity-averaged with 95% bootstrap CI. best in column.

Defense	T0	J85	J75	T2	T3	T5
clean	[TODO:a1]	[TODO:a2]	[TODO:a3]	[TODO:a4]	[TODO:a5]	[TODO:a6]
random	[TODO:b1]	[TODO:b2]	[TODO:b3]	[TODO:b4]	[TODO:b5]	[TODO:b6]
FSMG	[TODO:c1]	[TODO:c2]	[TODO:c3]	[TODO:c4]	[TODO:c5]	[TODO:c6]
ASPL-lite	[TODO:d1]	[TODO:d2]	[TODO:d3]	[TODO:d4]	[TODO:d5]	[TODO:d6]

Table 3: Stage B (DreamBooth-LoRA, 12-identity subset): mean identity similarity per defense and tier.

Defense	T0	J85	T2	T5
clean	[TODO:e1]	[TODO:e2]	[TODO:e3]	[TODO:e4]
random	[TODO:f1]	[TODO:f2]	[TODO:f3]	[TODO:f4]
FSMG	[TODO:g1]	[TODO:g2]	[TODO:g3]	[TODO:g4]

noticeable fraction of the T0 gap, JPEG-85 (the default quality of several major platforms) roughly halves it for fine-tuning-targeted defenses, and by the browser tier (T2) the remaining gap is a small fraction of its T0 value. Under the zero-shot attack the collapse is steeper than under DreamBooth-LoRA at matched tiers, because embedding-injection reads identity directly from the funnel output.

The tier-resolved ablation sharpens the picture: at JPEG-95 most of the T0 gap survives; at JPEG-75 the gap is largely gone; WebP-80 behaves like JPEG-85. DPR-2 browser rendering (T2) is no more destructive than DPR-1 *for the attack*, but is disproportionately destructive for protections, whose high-frequency structure interacts with resampling. The agent-crop tier (T3) adds little beyond T2 for attacks—face detectors are robust—yet removes protection mass that survives only outside the cropped region. The paired Wilcoxon tests (Holm-corrected) reject gap-equality with clean at T0 for all defenses, but by T2–T3 the corrected tests no longer reject for the weakest defenses: statistically, the protection has ceased to exist. [TODO:Insert exact mean/CI/p-values.]

5.2 F2: Zero-Shot Attacks Bypass Fine-Tuning Poisons

Stage B (Table 3) measures the classic target of these defenses: DreamBooth-LoRA fine-tuning on tiered galleries. At T0 the protections achieve their advertised effect. But the zero-shot family in Stage A reaches comparable identity extraction from the *same tiered galleries without any fine-tuning*: by T2–T4 the zero-shot extractor's similarity is already within a small margin of the fine-tuned extractor's, and it never touches the poisoned training dynamics the defenses were designed to corrupt. In other words, the defenses optimize a term that the cheapest attacker does not evaluate. [TODO:Insert Stage B table and zero-vs-tuned comparison.]

5.3 F3: Adaptive Tiers Close the Residual Gap

Stage C (Table 4) shows that even the residual protection visible at T0 is fragile against an attacker who adapts at the data level. Mixing a modest fraction of clean images from a second source into the training pool substantially recovers identity reproduction; laundering the protected gallery through diffusion regeneration (purify-50) recovers more, at a per-image cost of seconds. Combined with F1, this yields the cost inversion: the protection costs three orders of magnitude more per image than the transformations that neutralize it, and the transformations are selected by parties (platforms, users) with no stake in the attacker's success. [TODO:Insert Stage C table and cost ledger.]

Table 4: Stage C (adaptive, 12-identity subset): DreamBooth-LoRA identity similarity under clean-source mixing and purification.

Condition	mix-20	mix-50	purify-50
FSMG	[TODO:h1]	[TODO:h2]	[TODO:h3]

5.4 Imperceptibility and Detection-Failure View

Protection quality is only meaningful relative to image quality: FSMG-style perturbations sit at PSNR [TODO:i1] dB against the ϵ ball at 8/255, so the visibility budget is respected by construction. Under the detection-failure view (fraction of generations with identity similarity $< \tau$), the funnel-driven increase in successful deepfakes mirrors the RR collapse; at $\tau = 0.25$ and 0.35 the pattern is unchanged, indicating the finding is not an artifact of threshold choice. [TODO:Insert P@τ table.]

6 DISCUSSION: WHAT SURVIVES THE FUNNEL

6.1 Why Perturbation-Based Protection Is Structurally Fragile

The collapse we measure is not a collection of implementation artifacts; it follows from what protective perturbations are. A perturbation is a low-amplitude, typically high-frequency signal tuned to a specific computational path—the gradient of a fine-tuning loss through a particular denoiser. The funnel's platform tiers were never designed to preserve such signals: transcoders optimize rate–distortion, browsers resample for display, agents crop for relevance. Each operation is individually benign for semantic content and individually lethal for the perturbation's supporting structure, and their composition multiplies. Meanwhile the zero-shot attack family removes the very surface the perturbation poisons: IP-Adapter-FaceID reads an identity embedding off the funnel output and injects it through cross-attention, so there is no fine-tuning gradient to corrupt. A defense family whose mechanism is gradient poisoning and whose robustness story is adversarial training over hand-picked transformations cannot, in principle, certify itself against a pipeline it does not control. Recent limits-focused analyses reach consonant conclusions from the laboratory side [**?**]; our contribution is to show that no laboratory attacker is needed—the commodity web is adversary enough.

6.2 A Reporting Standard: RR@T

We propose that protection papers report, alongside T0 numbers: (i) the collapse curve RR(T) over the public funnel tiers, (ii) the tier at

which the Holm-corrected gap first becomes statistically indistinguishable from zero, and (iii) the cost ledger ($c_{\text{prot}}, c_{\text{funnel}}, \kappa$). This costs a page and changes the field's incentives: a defense that is strong only at T0 can no longer headline a single number. The tiers are intentionally implementable with commodity libraries so that adoption is a scripting exercise, not a research project; our release ships the registry.

6.3 The Durable Fix: Consent Provenance, Not Stronger Noise

If F1–F3 hold, the marginal return on stronger perturbations is small: the funnel is free and adaptive tiers launder what survives. The alternative is to change what is being protected. A machine-readable consent manifest—`face-consent.txt` served beside the image, declaring identity, scope (train/generate/none), and revocation—converts face protection from a pixel-level arms race into a provenance problem that platforms can enforce at ingestion and scrapers can be required to honor. This reframing has three advantages: it survives the funnel by construction (consent is metadata, not pixels); it composes with emerging content-credential infrastructure rather than fighting it; and it gives benign ecosystem actors—platforms, model trainers, inference providers—a technically checkable obligation, which perturbations never provide. We view ScrapeBench's role here as motivational: by quantifying exactly how little of today's pixel-level protection survives acquisition, the funnel makes the case for structural remedies concrete.

6.4 Limitations

Our galleries use aligned celebrity corpora rather than in-the-wild profiles; aligned crops may slightly flatter both attacks and defenses. The funnel tiers are faithful reconstructions, not captured traces from specific platforms; platform-internal pipelines vary and may be harsher or gentler than our composites. We test representative defenses from the open literature rather than every published variant, and our zero-shot attacker uses one adapter family; other identity adapters (e.g., InstantID-class [**?**]) may interact differently with T4–T5. Finally, ScrapeBench's adversary is resource-bounded and non-adaptive by design; a defense-aware red team with platform-grade infrastructure sits outside our threat model, and our bounds should be read as lower bounds on attacker capability.

7 CONCLUSION

We introduced ScrapeBench, an evaluation protocol that treats the web acquisition funnel—transcoding, browser and thumbnail rendering, agent capture, restoration, and diffusion-based regeneration—as the primary adversary of face anti-customization defenses. Measuring three representative defenses across two attack families under ten funnel tiers, we found that protections optimized for the laboratory's clean domain shed the large majority of their protection gap within the first platform-controlled tiers, that zero-shot identity injection extracts comparable identity without touching the fine-tuning gradients that perturbation defenses poison, and that modest data-level adaptation closes the remaining gap. Because funnel transformations cost orders of magnitude less than protection generation and are selected by parties with no stake in the attacker's success, the arms race they anchor is structurally unwinnable. ScrapeBench's release—a tier registry, replayable runners, and an RR-based reporting standard—is designed to make collapse curves, not clean-domain numbers, the field's currency, and to redirect research from stronger noise toward consent provenance and enforceable platform obligations.

REFERENCES

[] Zhen Chen, Yi Zhang, Xiangyu Yin, Chengxuan Qin, Xingyu Zhao, Xiaowei Huang, and Wenjie Ruan. 2025. Fragile by Design: On the Limits of Adversarial Defenses in Personalized Generation. arXiv:2511.10382. arXiv:2511.10382 http://arxiv.org/abs/2511.10382v1

[] Jiankang Deng, Jia Guo, Jing Yang, Niannan Xue, Irene Kotsia, and Stefanos Zafeiriou. 2018. ArcFace: Additive Angular Margin Loss for Deep Face Recognition. https://doi.org/10.1109/TPAMI.2021.3087709 arXiv:1801.07698

[] Tanjim Rahaman Fardin, S M Zunaid Alam, Mahadi Hasan Fahim, and Md Faysal Mahfuz. 2026. MetaCloak-JPEG: JPEG-Robust Adversarial Perturbation for Preventing Unauthorized DreamBooth-Based Deepfake Generation. arXiv:2604.18537. arXiv:2604.18537 http://arxiv.org/abs/2604.18537v1

[] Shaopeng Fu, Fengxiang He, Yang Liu, Li Shen, and Dacheng Tao. 2022. Robust Unlearnable Examples: Protecting Data Against Adversarial Learning. arXiv:2203.14533. arXiv:2203.14533 http://arxiv.org/abs/2203.14533v1

[] Kefu Guo, Fengfan Zhou, Hefei Ling, Ping Li, and Hui Liu. 2024. Improving the JPEG-resistance of Adversarial Attacks on Face Recognition by Interpolation Smoothing. arXiv:2402.16586. arXiv:2402.16586 http://arxiv.org/abs/2402.16586v1

[] Jiaming He, Jiashu Zhang, Guanyu Hou, Shuhan Ye, Hanwei Zhu, Yi Yu, and Xudong Jiang. 2026. ChronoLock: Protecting Videos from Unauthorized Text-to-Video Personalization. arXiv:2606.21146. arXiv:2606.21146 http://arxiv.org/abs/2606.21146v1

[] Hanxun Huang, Xingjun Ma, Sarah Monazam Erfani, James Bailey, and Yisen Wang. 2021. Unlearnable Examples: Making Personal Data Unexploitable. arXiv:2101.04898. arXiv:2101.04898 http://arxiv.org/abs/2101.04898v2

[] Yuxin Huang, Ziming Hong, Mingming Gong, Wanyu Wang, Jing Zhang, and Tongliang Liu. 2026. Delving into the Temporal Challenges of Unified Video Protection Against Image-to-Video and Fine-Tuning-based Customization. arXiv:2607.13336. arXiv:2607.13336 http://arxiv.org/abs/2607.13336v1

[] Jun Jia, Hongyi Miao, Yingjie Zhou, Linhan Cao, Yanwei Jiang, Wangqiu Zhou, Dandan Zhu, Hua Yang, Wei Sun, Xiongkuo Min, and Guangtao Zhai. 2025. DLADiff: A Dual-Layer Defense Framework against Fine-Tuning and Zero-Shot Customization of Diffusion Models. arXiv:2511.19910. arXiv:2511.19910 http://arxiv.org/abs/2511.19910v1

[] Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. 2017. Progressive Growing of GANs for Improved Quality, Stability, and Variation. arXiv:1710.10196. arXiv:1710.10196 http://arxiv.org/abs/1710.10196v3

[] Jungkon Kim, Cheolseung Jung, Jong-Min Choi, and Juseong Lee. 2026. Phantom: A Unified Face-Swap Deepfake Protection Framework with Latent and Spatial Constraints. arXiv:2606.31703. arXiv:2606.31703 http://arxiv.org/abs/2606.31703v1

[] Thanh Van Le, Hao Phung, Thuan Hoang Nguyen, Quan Dao, Ngoc Tran, and Anh Tran. 2023. Anti-DreamBooth: Protecting users from personalized text-to-image synthesis. arXiv:2303.15433. arXiv:2303.15433 http://arxiv.org/abs/2303.15433v2

[] Chanhui Lee, Donggyu Choi, Seunghyun Shin, Hae-Gon Jeon, and Jeany Son. 2026. Universal Image Immunization against Diffusion-based Image Editing via Semantic Injection. arXiv:2602.14679. arXiv:2602.14679 http://arxiv.org/abs/2602.14679v2

[] Ang Li, Yichuan Mo, Mingjie Li, and Yisen Wang. 2024. PID: Prompt-Independent Data Protection Against Latent Diffusion Models. arXiv:2406.15305. arXiv:2406.15305 http://arxiv.org/abs/2406.15305v1

[] Jiahao Li, Yiqiang Chen, Yunbing Xing, Yang Gu, and Xiangyuan Lan. 2025. A Survey on Unlearnable Data. arXiv:2503.23536. arXiv:2503.23536 http://arxiv.org/abs/2503.23536v3

[] Yixin Liu, Chenrui Fan, Yutong Dai, Xun Chen, Pan Zhou, and Lichao Sun. 2023. MetaCloak: Preventing Unauthorized Subject-driven Text-to-image Diffusion-based Synthesis via Meta-learning. arXiv:2311.13127. arXiv:2311.13127 http://arxiv.org/abs/2311.13127v5

[] Hengrui Lou, Zunlei Feng, Jinsong Geng, Erteng Liu, Lechao Cheng, Jie Lei, Jie Song, Mingli Song, and Yijun Bei. 2025. A Large-scale Universal Evaluation Benchmark For Face Forgery Detection. In *Proceedings of the 33rd ACM International Conference on Multimedia*. ACM, 12592–12599.

[] Takuto Onikubo and Yusuke Matsui. 2024. High-Frequency Anti-DreamBooth: Robust Defense against Personalized Image Synthesis. arXiv:2409.08167. arXiv:2409.08167 http://arxiv.org/abs/2409.08167v3

[] Tarik Can Ozden, Ozgur Kara, Oguzhan Akcin, Kerem Zaman, Shashank Srivastava, Sandeep P. Chinchali, and James M. Rehg. 2024. DiffVax: Optimization-Free Image Immunization Against Diffusion-Based Editing. arXiv:2411.17957. arXiv:2411.17957 http://arxiv.org/abs/2411.17957v2

[] Gan Pei, Jiangning Zhang, Menghan Hu, Zhenyu Zhang, Chengjie Wang, Yunsheng Wu, Guangtao Zhai, Jian Yang, and Dacheng Tao. 2026. Deepfake Generation and Detection: A Benchmark and Survey. *Comput. Surveys* 58, 11 (2026), 1–41. https://doi.org/10.1145/3801962

[] Sen Peng, Mingyue Wang, Jianfei He, Jijia Yang, and Xiaohua Jia. 2025. CAT: Contrastive Adversarial Training for Evaluating the Robustness of Protective Perturbations in Latent Diffusion Models. arXiv:2502.07225. arXiv:2502.07225 http://arxiv.org/abs/2502.07225v2

[] Tianrui Qin, Xitong Gao, Juanjuan Zhao, Kejiang Ye, and Cheng-Zhong Xu. 2023. Learning the Unlearnable: Adversarial Augmentations Suppress Unlearnable Example Attacks. arXiv:2303.15127. arXiv:2303.15127 http://arxiv.org/abs/2303.15127v1

[] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2021. High-Resolution Image Synthesis with Latent Diffusion Models. arXiv:2112.10752. arXiv:2112.10752 http://arxiv.org/abs/2112.10752v2

[] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. 2022. DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation. arXiv:2208.12242. arXiv:2208.12242 http://arxiv.org/abs/2208.12242v2

[] Hadi Salman, Alaa Khaddaj, Guillaume Leclerc, Andrew Ilyas, and Aleksander Madry. 2023. Raising the Cost of Malicious AI-Powered Image Editing. arXiv:2302.06588. arXiv:2302.06588 http://arxiv.org/abs/2302.06588v1

[] Pedro Sandoval-Segura, Jonas Geiping, and Tom Goldstein. 2023. JPEG Compressed Images Can Bypass Protections Against AI Editing. arXiv:2304.02234. arXiv:2304.02234 http://arxiv.org/abs/2304.02234v2

[] Pedro Sandoval-Segura, Vasu Singla, Jonas Geiping, Micah Goldblum, and Tom Goldstein. 2023. What Can We Learn from Unlearnable Datasets? arXiv:2305.19254. arXiv:2305.19254 http://arxiv.org/abs/2305.19254v3

[] Pedro Sandoval-Segura, Vasu Singla, Jonas Geiping, Micah Goldblum, Tom Goldstein, and David W. Jacobs. 2022. Autoregressive Perturbations for Data Poisoning. arXiv:2206.03693. arXiv:2206.03693 http://arxiv.org/abs/2206.03693v3

[] Shawn Shan, Emily Wenger, Jiayun Zhang, Huiying Li, Haitao Zheng, and Ben Y. Zhao. 2020. Fawkes: Protecting Privacy against Unauthorized Deep Learning Models. USENIX Security Symposium 2020. arXiv:2002.08327 http://arxiv.org/abs/2002.08327v2

[] Yiren Song, Pei Yang, Hai Ci, and Mike Zheng Shou. 2024. IDProtector: An Adversarial Noise Encoder to Protect Against ID-Preserving Image Generation. arXiv:2412.11638. arXiv:2412.11638 http://arxiv.org/abs/2412.11638v2

[] Matteo Trippodo, Federico Becattini, and Lorenzo Seidenari. 2025. Immunizing Images from Text to Image Editing via Adversarial Cross-Attention. arXiv:2509.10359. arXiv:2509.10359 http://arxiv.org/abs/2509.10359v1

[] Guanyu Wang, Kailong Wang, Yihao Huang, Mingyi Zhou, Geguang Pu, and Li Li. 2025. Privacy Protection Against Personalized Text-to-Image Synthesis via Cross-image Consistency Constraints. arXiv:2504.12747. arXiv:2504.12747 http://arxiv.org/abs/2504.12747v3

[] Qixun Wang, Xu Bai, Haofan Wang, Zekui Qin, Anthony Chen, Huaxia Li, Xu Tang, and Yao Hu. 2024. InstantID: Zero-shot Identity-Preserving Generation in Seconds. arXiv:2401.07519. arXiv:2401.07519 http://arxiv.org/abs/2401.07519v2

[] Tao Wang, Yushu Zhang, Xiangli Xiao, Kun Xu, Lin Yuan, Wenying Wen, and Yuming Fang. 2025. Make Identity Unextractable yet Perceptible: Synthesis-Based Privacy Protection for Subject Faces in Photos. arXiv:2509.11249. arXiv:2509.11249 http://arxiv.org/abs/2509.11249v1

[] Hu Ye, Jun Zhang, Sibo Liu, Xiao Han, and Wei Yang. 2023. IP-Adapter: Text Compatible Image Prompt Adapter for Text-to-Image Diffusion Models. arXiv:2308.06721. arXiv:2308.06721 http://arxiv.org/abs/2308.06721v1

[] Boyang Zheng, Chumeng Liang, and Xiaoyu Wu. 2023. Targeted Attack Improves Protection against Unauthorized Diffusion Customization. arXiv:2310.04687. arXiv:2310.04687 http://arxiv.org/abs/2310.04687v5

A APPENDIX

A.1 Reproducibility

The complete harness (`scrapebench`) is released under an open license with: (i) the tier registry implementing T0–T6 as pure image functions with deterministic fallbacks; (ii) per-stage runners (`protection_run`, `stage_a`, `stage_b`, `stage_c`) writing append-only CSVs keyed by (stage, identity, defense, tier, attack) and resuming deterministically; (iii) the report generator producing collapse curves, retention tables, paired Wilcoxon tests with Holm correction, Cliff's δ, and bootstrap CIs; (iv) a cost ledger emitted per cell. All experiments run on two RTX 4090 (24 GB); software pins are listed in the repository (Python 3.10, torch 2.x CUDA, diffusers 0.31, transformers 4.44, peft 0.13, insightface 0.7). Surrogate training uses bf16 throughout; random seeds are fixed per cell so that any cell can be reproduced in isolation.

A.2 Ethics Statement

This work evaluates *defenses*, not attack tooling: the attack pipelines used are publicly released research artifacts (DreamBooth-LoRA [**?**]; IP-Adapter-FaceID [**?**]) executed with default settings on a public celebrity face corpus under its license, and no attempt is made to identify or de-anonymize the depicted individuals. Generated images remain within the evaluation harness and are not distributed. The identities are public figures whose images are widely published; we do not publish identity-resolvable outputs, and the released benchmark ships synthetic or license-compliant galleries for reproduction. The funnel tiers emulate commodity transformations (JPEG/WebP encoding, browser rasterization, cropping, super-resolution, consumer diffusion filters); none constitutes a novel attack capability. We hope the measured collapse curves inform platform policy: our results argue for consent manifests and ingestion-time obligations rather than stronger pixel noise, which is precisely the non-adversarial, harm-reducing direction. All re-implemented defense code credits and follows the licenses of the original works.

A.3 Additional Results

Full per-tier tables for Stage A (including JPEG-95, JPEG-65, WebP-80, DPR-2, SR, img2img-50, and inversion tiers), the detection-failure view at thresholds $\tau \in \{0.25, 0.30, 0.35\}$, per-identity collapse curves, quality metrics (PSNR/LPIPS) of every protection under every tier, and the complete cost ledger are included in the released result logs. [TODO:Appendix tables will be generated by report.py from the final runs.]